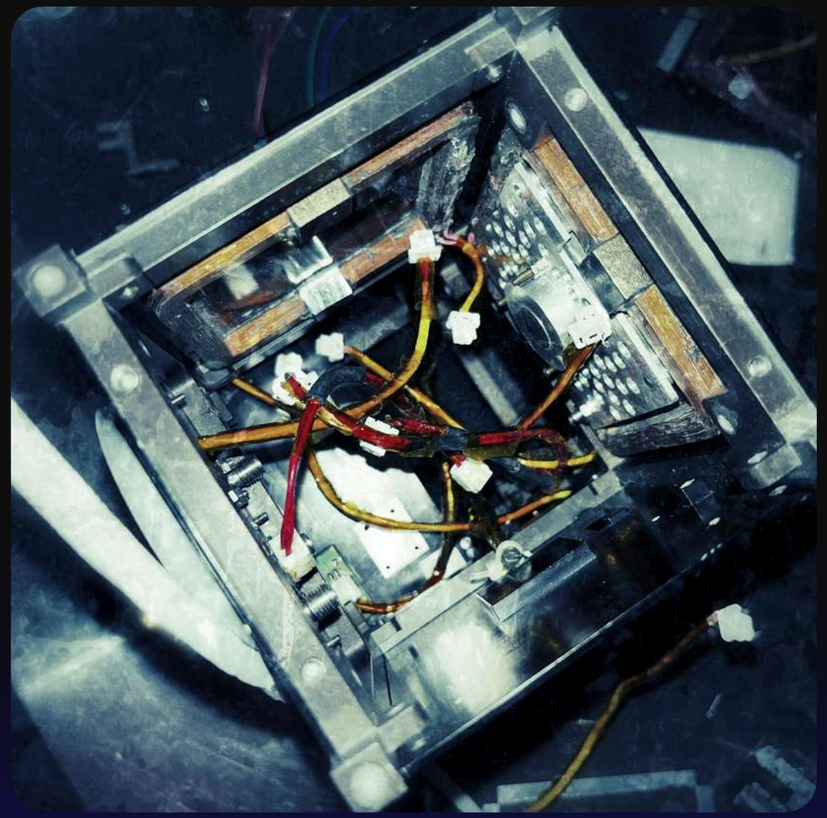
MANIPAL
ACADEMY of HIGHER EDUCATION
(Deemed to be University under Section 3 of the UGC Act, 1956)

SUBSYSTEMS
TEAM STRUCTURE

Parikshit

STUDENT SATELLITE

ABOUT PARIKSHIT



Parikshit is a student-led satellite development initiative associated with Manipal Institute of Technology. The project provides students with practical exposure to the design, development and integration of a real spacecraft. The project focuses on the development of a 2U-class nanosatellite intended for Polar Low Earth Orbit (LEO).

Mission Objectives

- Perform thermal Earth imaging using an onboard thermal camera.
- Demonstrate an electrodynamic tether as an experimental deorbiting mechanism.
- Develop and integrate spacecraft subsystems completely through multidisciplinary student teams.
- Provide hands-on experience in aerospace engineering, electronics, embedded systems, software, communication and project management.

Key Idea

Parikshit is not a single engineering project, it is an integrated system in which multiple specialized subsystems must work together to accomplish one mission.

Satellite Subsystems

Parikshit is divided into seven primary functional areas:

- ADCS — Attitude Determination & Control
- COMMS — Communications & Ground Station
- EPS — Electrical & Power
- ODHS — Onboard Data Handling
- Payload — Mission-specific equipment
- STMS — Structures, Thermal & Mechanics
- Administration & Operations

Each subsystem has a specific responsibility while working with the others.

ADCS & EPS



ADCS -Attitude Determination & Control

Purpose: Controls the satellite’s orientation.

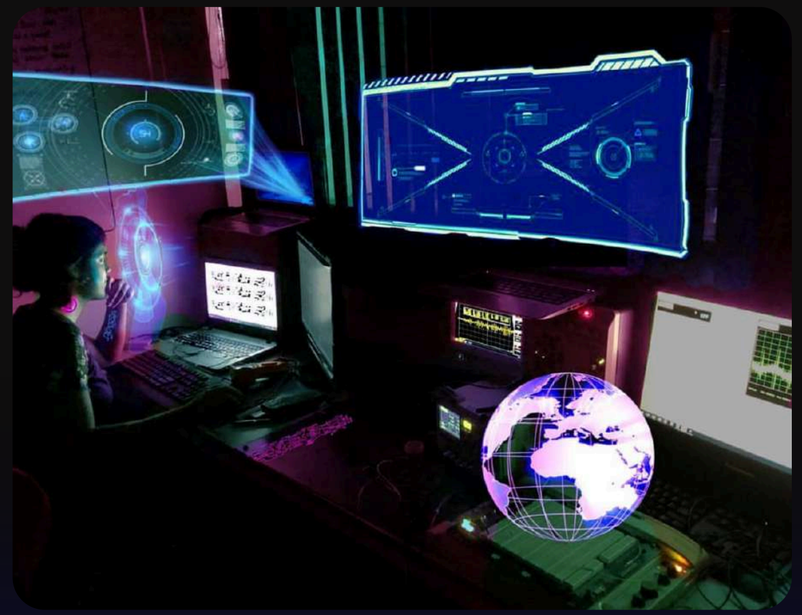
- Determines spacecraft attitude using sensors.
- Maintains required orientation for mission operations.
- Uses sun sensors, magnetometers and IMU.
- Uses magnetorquers for attitude control.

EPS - Electrical & Power

Purpose: Provides and manages electrical power.

- Solar panels generate power.
- Batteries store energy.
- Regulates and distributes power to subsystems.
- Provides electrical protection and power management.

ODHS & COMMS



ODHS - Onboard Data Handling

Purpose: Manages spacecraft data and commands.

- Processes commands.
- Collects and stores telemetry.
- Monitors spacecraft health.
- Interfaces with other subsystems.

COMMS -Communications

Purpose: Connects the satellite with the ground station.

- Receives commands from Earth (uplink).
- Sends telemetry and mission data to Earth (downlink).
- Uses communication hardware such as transceivers and antennas.

PAYLOAD



The payload performs the actual mission objectives of the satellite.

Primary Payload

Thermal Camera

- Used for thermal imaging of Earth.
- Generates mission-specific imaging data.

Secondary Payload

Electrodynamic Tether

- Experimental technology for spacecraft deorbiting.
- Uses interaction with Earth's magnetic field.

Main Responsibilities

- Data acquisition
- Experiment scheduling
- Payload health monitoring
- Data transfer to ODHS

STMS



Structures, Thermal & Mechanics

STMS provides the spacecraft’s physical and thermal support.

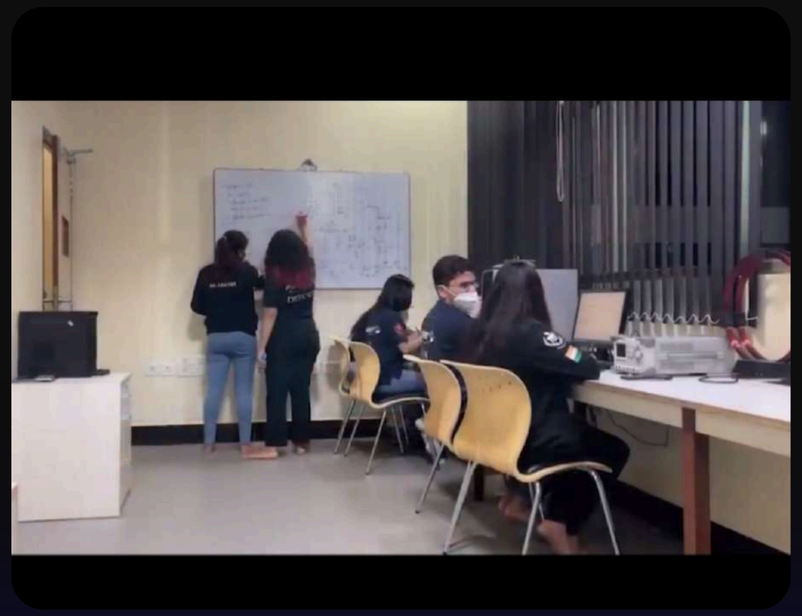
Responsibilities

- Designs the satellite’s structural framework.
- Ensures the spacecraft can withstand launch loads and vibration.
- Manages the thermal environment.
- Supports deployment mechanisms.
- Provides mounting for components such as solar panels.

Why is it important?

STMS creates the physical environment in which all other spacecraft systems operate.

Administration & Operations



A satellite project requires more than engineering.

Administration & Operations handles:

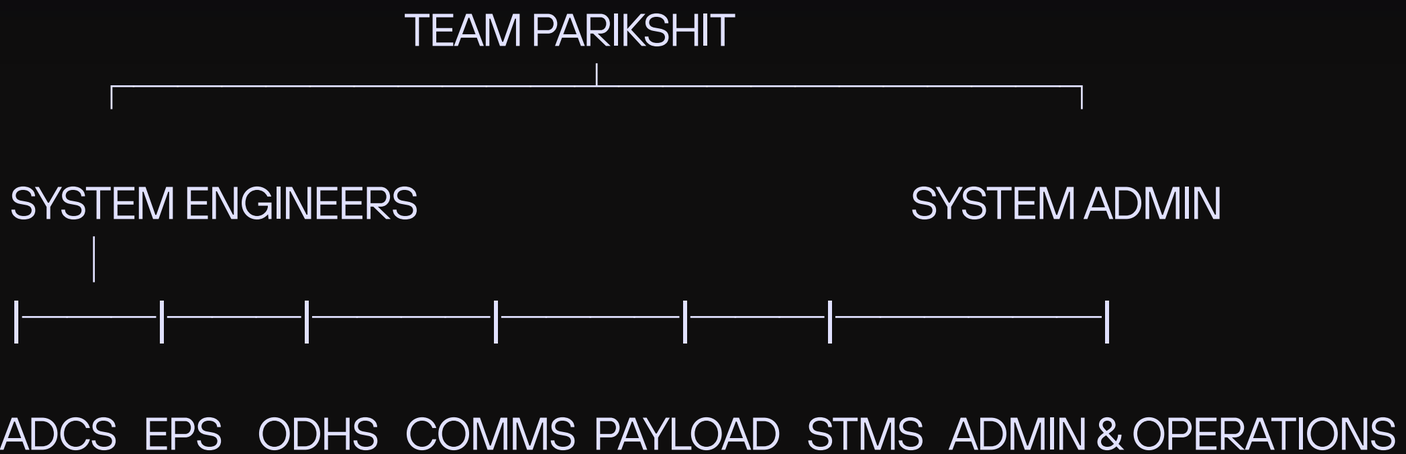
- Sponsor and company outreach
- Website and social media
- Graphic design
- Permissions and documentation
- External communication
- Coordination with universities, agencies and sponsors

Key Point

Successful engineering projects require both technical development and effective management.

Team Structure

How Team Parikshit Works





- Each subsystem has specialized team members.
- System-level leadership coordinates the technical teams.
- Administrative members support outreach and operations.
- Faculty guidance supports the overall team.

How Everything Connects

Subsystem Integration

EPS → Provides Power
↓
ODHS → Processes & Manages Data
↓
ADCS → Controls Orientation
↓
PAYLOAD → Performs Mission
↓
ODHS → Stores Data
↓
COMMS → Sends Data to Ground

What I Learned

- Satellite development requires multiple specialized teams.
- Every subsystem depends on the others.
- Good system integration and teamwork are essential for mission success.

REFERENCES

- Parikshit Student Satellite: https://parikshitspace.in/
- Subsystems: https://parikshitspace.in/subsystems
- Team: https://parikshitspace.in/team

thank you.

